Academic year 2025-2026
Course : Optimization

Assignment 8

Exercise 1. ***(3D data set fitting with linear least squares)***
Polynomials in two variables are functions $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ consisting of the sum of terms of the form $b\, x_1^n x_2^m$, where $b \in \mathbb{R}$ is called a ***coefficient****. The degree of each term $b\, x_1^n x_2^m$ is the sum of the exponents $n+m$, and the degree of the polynomial is the largest degree among all the terms of the polynomial. Examples :*

$$x_1^2 x_2 - 6x_1^3 x_2^{12} + 10x_1^2 - 7x_2 + 1 \qquad \textit{degree} \quad 15$$
$$x_1^4 x_2^2 - x_1^3 x_2^3 - x_1 x_2 + x_1^4 \qquad \textit{degree} \quad 6$$

The aim of this exercise is to implement a 3D data set fitting method using a polynomial regression by minimizing the linear least squares

$$\underset{\beta \in \mathbb{R}^p}{\arg\min} \|y - X\beta\|^2 \tag{0.1}$$

where the vector β contains the coefficients of the polynomial.

The dataset $\{(x_{1i}, x_{2i}, y_i)\}$ considered in this exercise has been generated as random points from a true function (see Figure 0.1). All the data are available in the file ***ex1.py****.*

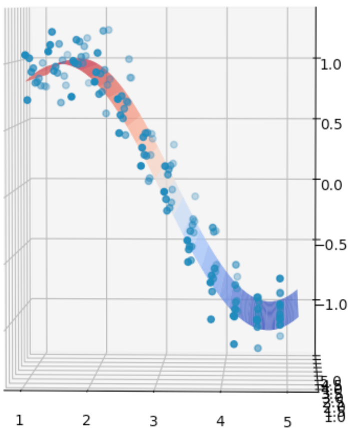

FIGURE 0.1: Dataset and the true function from which it has been generated.

1. Determine p in (0.1) in function of the degree of the polynomial.

2. Write a Python function that, given a degree, returns the corresponding matrix X in (0.1) for the given dataset $\{(x_{1i}, x_{2i}, y_i)\}$.

3. Using the matrix X constructed in 2., compute the solution of the minimization problem (0.1) for polynomials of degrees 1,3,4,5,6,7. For each degree, plot the surface generated by the polynomial, plot the dataset and the true function as well, to show how the polynomial adjusts to the data and the true function.

4. Compute the mean square error of each polynomial with respect to both the dataset and the true function. Which polynomial best approximates the true function ?

Exercise 2. ***(3D data set fitting with nonlinear least squares)***
Given the dataset $\{(x_{1i}, x_{2i}, y_i)\}$ *of Exercise 1, the aim of this Exercise is to determine the coefficients* $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)^T$ *of the model*

$$f(\beta; x) = \beta_0 + \beta_1 \sin(\beta_2 x_1) \cos(\beta_3 x_2)$$

solving the nonlinear least squares problem

$$\arg\min_{\beta} \sum_{i=1}^{m} r_i(\beta)^2$$

where $r_i(\beta) = f(\beta; x_{1i}, x_{2i}) - y_i$ *are the components of the residual vector*

$$r(\beta) = \begin{pmatrix} r_1(\beta) \\ r_2(\beta) \\ \vdots \\ r_m(\beta) \end{pmatrix}$$

1. Write a Python function that evaluates the model $f(\beta; x)$*.*

2. Write a Python function that evaluates the residual $r(\beta)$ *for the dataset of Exercise 1.*

3. Write a Python function that computes the value of the Jacobian of $r(\beta)$*, where the partial derivatives have been determined in an analytic manner.*

4. Write a Python function that implements the Levenberg-Marquardt algorithm (Algoritmo 1 in the class of 04/23/2026) to solve the non linear least squares problem.

5. Test the algorithm with the following parameters :
- Initial point $\beta_0 = (1, 1, 0.75, 0.5)^T$
- $\mu_{ref} = 0.001$
- Threshold $\tau = \sqrt{m}\,\epsilon_m^{1/3}$*, where* ϵ_m *is the machine espilon.*
- Maximum number of iterations $N = 200$*.*

(i) Print the final vector β_k*, the final value* $f(x_k)$*, the final number of iteraciones* k *and the variable res that indicates whether the threshold has been reached before the end of the iterations.*

(ii) Calculate the mean square error with respect to the dataset and the true function, and print the values.

(iii) Plot the surface generated by the model, plot the dataset and the true function as well, to show how the model adjusts to the data and the true function.

6. Repeat 5. with the following initial point $\beta_0 = (1, 0.5, 0.75, 0.5)^T$*.*

Exercise 3. ***Constrained least-squares***

Let $H \in \mathbb{R}^{p\times n}$ be matrix of rank n, and the function $f\colon \mathbb{R}^n \longrightarrow \mathbb{R}$ defined by

$$f(x) = \|y - Hx\|^2.$$

1. Let $C \in \mathbb{R}^{q\times n}$ and $b \in \mathbb{R}^q$. Show that f has an unique minimum on the set $S := \{x \in \mathbb{R}^n, Cx = b\}$.

2. Assuming that C is such that the matrix $C(H^T H)^{-1} C^T$ has rank q, determine the minimum of f on S.

3. Express the minimum of f on S in function of the minimum of f on $\mathbb{R}^n$.

Exercise 4. ***Projection onto convex sets***

Using KKT conditions, determine the projection operator onto :
(i) the closed ball of center (0,0) and radius R
(ii) a closed box $[a_1, b_1] \times [a_2, b_2]$, where $a_1 < b_1$ and $a_2 < b_2$.